The Gaussian equivalence property and feature learning in neural networks

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Abstract

We consider the problem of feature learning in two-layer neural networks trained with one large step of gradient descent. Our analysis accommodates spiked input data where the bulk is allowed to be anisotropic and ill-conditioned, as well as regularization in the form of weight decay. We derive a decomposition of the trained feature matrix into a sum of bulk and spike terms, which is accomplished using generalized Hermite polynomials. One of our key findings is that the number of potential spikes that emerge from the bulk spectrum of the feature matrix increases polynomially as the step size grows. We also identify a threshold parameter (which relates the data spike size, the network scaling, and the step size) that governs feature learning. We state a data-adaptive, resolvent-level Gaussian equivalence property for the untrained feature matrix and outline the arguments used to prove it. Using this result, we precisely characterize the test loss of the ridge regression estimator using the learned feature matrix. This report is based on ongoing work of the second author, Rishi Sonthalia, and Guido Montúfar.

1 Introduction

Neural networks have achieved remarkable performance on a wide range of tasks in numerous domains. It is widely believed that the key driver of this success is the ability of neural networks to *learn nonlinear features* from data (Donahue et al., 2017, Yang and Hu, 2021, Shi et al., 2022, Radhakrishnan et al., 2024, Moniri et al., 2024). However, while much work has been done to explain the mechanisms underlying feature learning, a comprehensive theoretical understanding of this phenomenon remains out of reach. The present work sits at the intersection of three lines of research: 1) *random features and kernel asymptotics* (Hu et al., 2024), 2) *feature learning in shallow neural networks* (Ba et al., 2022), and 3) *spiked or anisotropic covariance models* (Wang et al., 2024).

In recent years, many results in the study of nonlinear feature learning in shallow neural networks have shown that taking one large gradient step yields nontrivial improvement in the prediction risk (Ba et al., 2022, Dandi et al., 2024, Moniri et al., 2024). These works consider the proportional asymptotic limit and use tools from high-dimensional probability and random matrix theory. In particular, a major ingredient used to prove results on this topic is *Gaussian equivalence* (Montanari and Saeed, 2022, Wen et al., 2025), a theoretical framework stating that the behavior of high-dimensional, nonlinear features can be captured by linear Gaussian surrogates.

We aim to address the following question: *how do we characterize the asymptotic generalization error under more general conditions of anisotropy and ill-conditioning?* In short, we extend the analysis in Moniri et al. (2024) to the setting of Sonthalia et al. (2025). Utilizing generalized Hermite polynomials, we decompose the feature matrix into a sum of bulk and spike terms. We then discuss a Gaussian equivalence-type property for the bulk term; we will also highlight some of the arguments used to prove this result. Applying this property to the untrained features then makes our analysis of the test error more tractable.

2 Problem setup

We work in the proportional asymptotic regime

$$\frac{n}{d} \to \psi_1 \in (0,1), \quad \frac{m}{d} \to \psi_2 \in \mathbb{R}_{>0}, \quad d \to \infty,$$

where n is the sample size, d is the input dimension, and m is the hidden layer width. This reflects the setting where the network width and the data size are comparable. We consider a two-layer network $f\colon \mathbb{R}^d \to \mathbb{R}$ of the form

$$f(x) = \gamma_m \, a^\top \sigma(Wx),$$

with inner-layer weights $W = [w_1 \quad \dots \quad w_m]^\top \in \mathbb{R}^{m \times d}$ whose rows are initialized satisfying $w_j \in S^{d-1}$, outer weights $a \in \{\pm 1\}^m$ sampled i.i.d. and then fixed, and width-dependent scale γ_m. The parameter γ_m defines the two common scaling regimes:

- Neural tangent kernel (NTK): We set $\gamma_m = \Theta(1/\sqrt{m})$. This is associated with lazy training where the inner-layer weights vary little (Chizat et al., 2019, Jacot et al., 2018) and yields the output variance $\mathrm{Var}(f(x)) = \Theta(1)$.
- Mean field (MF): We set $\gamma_m = \Theta(1/m)$. This is associated with feature learning (Chizat and Bach, 2018, Mei et al., 2018) and yields the output variance $\mathrm{Var}(f(x)) = o(1)$.

The training inputs $x_1, \dots, x_n \in \mathbb{R}^d$ are i.i.d. Gaussian with spiked covariance

$$x_i \sim \mathcal{N}(0, \Sigma), \quad \Sigma = \hat{\Sigma} + \zeta^2 q q^\top, \quad \zeta = n^\nu, \quad \nu \geq 0,$$

where $q \in \mathbb{S}^{d-1}$ is the spike direction and the bulk covariance $\hat{\Sigma} \in \mathbb{R}^{d \times d}$ has eigenvalues

$$\lambda_k(\hat{\Sigma}) = k^{-\alpha}, \quad k = 1, \dots, d, \quad \alpha \geq 0.$$

Equivalently, if $X \in \mathbb{R}^{n \times d}$ stores the samples row-wise, then

$$X = X_B + X_S = X_B + \zeta z q^\top,$$

where the rows of X_B are i.i.d. $\mathcal{N}(0, \hat{\Sigma})$ and $z \sim \mathcal{N}(0, I_n)$. This bulk covariance allows us to study more general data models such as those following power-law distributions (Clauset et al., 2009). We also define $\tilde{X} = \tilde{X}_B + \tilde{X}_S \in \mathbb{R}^{n \times d}$ to be the new training data (used to update the feature matrix). This is to ensure independence in our analysis. Given a differentiable loss $\ell\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, we define the residue vector

$$r = \left[\frac{\partial \ell(f(x_1), y_1)}{\partial f(x_1)}, \dots, \frac{\partial \ell(f(x_n), y_n)}{\partial f(x_n)} \right]^\top \in \mathbb{R}^n.$$

We interpret r as the component of the targets not captured by the model predictions. For each hidden unit, define the pre-activation variance

$$\tau_i^2 = w_i^\top \Sigma w_i,$$

so that $w_i^\top x \sim \mathcal{N}(0, \tau_i^2)$ when $x \sim \mathcal{N}(0, \Sigma)$. Finally, we consider the two-step training procedure introduced in the works of Damian et al. (2022) and Ba et al. (2022), which has also been studied in subsequent works (see, e.g., Dandi et al., 2024, Moniri et al., 2024). We first train the inner-layer weights W by taking a (large) gradient step, and then optimize the outer-layer weights a via ridge regression.

2.1 Assumptions

We make the following assumptions in order to present our theoretical results. There are other technical conditions that will be needed, which we will either briefly mention or suppress for brevity. We note that the setup of the paper is exactly Assumptions 1–3 from Sonthalia et al. (2025). We additionally need the following assumption from Sonthalia et al. (2025).

Assumption 1 (Activation function)**.** *The activation function $\sigma : \mathbb{R} \to \mathbb{R}$ satisfies:*

1. ***Smoothness:*** *σ' and σ'', first and second derivatives of σ, exist almost everywhere on $\mathbb{R}$.*
2. ***Lipschitzness:*** *σ and σ' are L-Lipschitz for some constant $L > 0$.*

3. ***Non-trivial expected derivative:*** *For $x \sim N(0, \Sigma)$ and W, let $\mu_j = \mathbb{E}_x[\sigma'(w_j^T x)]$ (expectation over x for a given w_j). We assume $\mu_j = \Omega(1)$ for all j.*

Commonly used activation functions such as Sigmoid, Tanh, ELU, Softplus, and Swish satisfy these conditions (under additional mild assumptions). In addition to the above, we shall need to bound the Hermite coefficients of the expansion of the activation function. However, since the inputs are $\mathcal{N}(0, \tau_i^2)$ we need to work with the generalized Hermite polynomials. For $\tau > 0$, let He_k^τ denote the degree-k generalized Hermite polynomial orthogonal in $L^2(\mathcal{N}(0, \tau^2))$.

Assumption 2 (Decay of generalized Hermite coefficients)**.** *For some constants $C, \omega > 0$, the coefficients satisfy $c_1^\tau \neq 0$ and*

$$(c_k^\tau)^2 k! \leq C k^{-3/2-\omega}, \quad k \geq 1.$$

To analyze the generalization error, we consider the following model for the teacher function, leading to a single-index model.

Assumption 3 (Single-index model)**.** *Let $f_*(x) : \mathbb{R}^d \to \mathbb{R}$ be $f_*(x) = \sigma_*(x^\top \beta_*)$ for all x, where $\beta_* \in \mathbb{R}^d$ is an unknown parameter with $\beta_* \sim \mathcal{N}\big(0, \frac{1}{d} I_d\big)$ and σ_* is an activation function satisfying Assumption 4 in Sonthalia et al. (2025).*

This condition follows from prior work (see Ba et al. (2022), Goldt et al. (2022), Cui et al. (2024) etc.) and is standard. Dandi et al. (2024) showed how shallow neural networks are only able to learn a single-index approximation of a multi-index model and Braun et al. (2025) studied the problem of learning single index models from anisotropic Gaussian inputs. We also assume the teacher activation admits the following expansion:

Assumption 4 (Hermite expansion of teacher activation)**.** *The teacher activation $\sigma_* : \mathbb{R} \to \mathbb{R}$ has the following generalized Hermite expansion in L^2:*

$$\sigma_*(z_*) = \sum_{\ell=1}^{\infty} c_{*,\ell}^{\tau_*} He_\ell^{\tau_*}(z), \quad c_{*,\ell}^{\tau_*} = \frac{1}{\ell! {\tau_*}^{2\ell}} \mathbb{E}[\sigma_*(z_*) He_\ell^{\tau_*}(z_*)],$$

where $z_* \sim \mathcal{N}(0, \tau_*^2)$.

3 Feature expansion

Prior work has worked on understanding the structure of the gradient (see, e.g., Ba et al., 2022, Sonthalia et al., 2025). However, knowing the structure of the gradient alone is not sufficient to fully understanding the presence of feature learning. In this section, we examine how the updated weight matrices affect the hidden features. Depending on the size of the spikes can result in multiple, polynomially many, spikes at the level of the features.

Our theory also accommodates the common regularization technique of ℓ_2 weight decay. Let λ_{wd} denote the regularization parameter. Adding $\frac{\lambda_{\text{wd}}}{2}\|W\|_F^2$ to the loss function yields the modified gradient $G^{(\lambda_{\text{wd}})} = G^{(0)} + \lambda_{\text{wd}} W$.

Recalling $\tau_i^2 = w_i^\top \Sigma w_i$, we set

$$F_1^{(L,\tau)} \coloneqq \begin{bmatrix} \sigma_L^{\tau_1}\left(\tilde{X} w_1^{(1)}\right) \\ \vdots \\ \sigma_L^{\tau_m}\left(\tilde{X} w_m^{(1)}\right) \end{bmatrix}, \quad F_0^{(L,\tau)} \coloneqq \begin{bmatrix} \sigma_L^{\tau_1}\left((1-\eta\lambda_{\text{wd}})\tilde{X}_B w_1^{(0)}\right) \\ \vdots \\ \sigma_L^{\tau_m}\left((1-\eta\lambda_{\text{wd}})\tilde{X}_B w_m^{(0)}\right) \end{bmatrix}, \tag{1}$$

where $\sigma_L^{\tau_j}$ is the L-term generalized Hermite expansion of σ with variance τ_j, and $w_i^{(0)}$ and $w_i^{(1)}$ are the i-th columns of the weight matrix at initialization and after one gradient step is applied, respectively. Using ℓ_2 weight decay means the pre-activations of $F_0^{(L,\tau)}$ are scaled by $(1 - \eta\lambda_{\text{wd}})$. We also define the spike term

$$S \coloneqq (1-\eta\lambda_{\text{wd}}) W_0 \tilde{X}_S^\top + \eta \hat{G} \tilde{X}^\top,$$

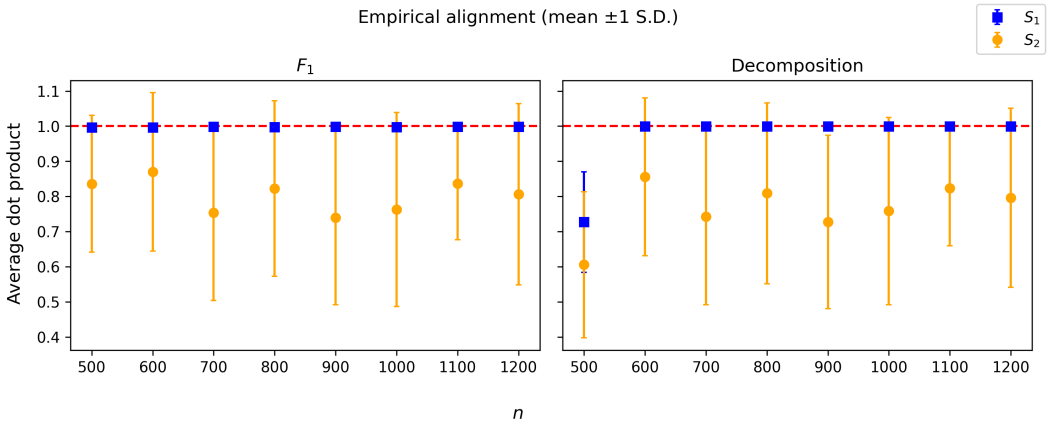


Figure 1: Empirical alignment (normalized inner product) of the top two singular vectors of $F_1^{(L,\tau)}$ (left) and its expansion using $S_1 + S_{12} + S_2$ (right) (as defined in Theorem 3.1) with the top singular vector of S_1 and S_2, respectively. We average over 20 samples. Parameters: $\psi_1 = 2/3$, $\psi_2 = 4/3$, $\alpha = 0$, $\nu = 1/8$, $\eta = m^{0.3}$, $\gamma_m = 1/m$, $\ell = 1$, $\lambda = 1/\eta$, tanh activation, and triple-index model targets.

where $\hat{G} = S_1 + S_{12} + S_2$ is the low-rank gradient proxy studied in Sonthalia et al. (2025). Each of these are rank-one matrices, which represent spike contributions from the residue r (S_1) and from the input data (S_2). The matrix S_{12} interpolates between the two, i.e., S_{12} and S_1 have the same right singular vector and S_{12} and S_2 have the same left singular vector.

Sonthalia et al. (2025) establishes that the rank of $\hat{G}$ depends on ν. In particular, the gradient is approximately rank one and aligns with the residue $S_1 + S_{12}$ if $\nu \in [0, 0.25)$, and is approximately the rank-two matrix $S_1 + S_{12} + S_2$ if $\nu \in [0.25, 0.5)$. The trained feature matrix also inherits this low-rank structure: under a suitable step size (which is defined precisely in Theorem 3.1) $F_1^{(L,\tau)}$ is rank one if $\nu = 0$, is rank two if $\nu \in (0, 0.25)$, and rank three if $\nu \in (0.25, 0.5)$. This is empirically verified in Figure 1.

Theorem 3.1 (Y.–Sonthalia–Montúfar)**.** *Let L be fixed and $\hat{\gamma} := \log_m\left(\frac{1}{\gamma_m}\right) - \frac{1}{2}$. Given step size $\eta \asymp m^{\delta}$, define $\kappa := \frac{1}{2} + \hat{\gamma} - \delta - 2\nu$. Suppose Assumption 2 holds.*

1. Subcritical spike regime. *Suppose $0 < \kappa < \frac{1}{2}$. Let $\ell \in \{1, \dots, L\}$ be the unique integer such that $\kappa \in \left(\frac{1}{2(\ell+1)}, \frac{1}{2\ell}\right)$. Assume in addition that $\kappa > \frac{1}{2} - \frac{1}{2L}$ and that λ_{wd} satisfies*

$$|1 - \eta\lambda_{wd}| = o\left(n^{-(L-1)(\frac{1}{2}-\kappa)} \log^{-\frac{3L-1}{2}} n\right).$$

Then, with probability $1 - o(1)$,

$$F_1^{(L,\tau)} = F_0^{(L,\tau)} + \sum_{k=1}^{\ell} (c_k^{(\tau)} \mathbf{1}_n^\top) \circ S^{\circ k} + o\left(\sqrt{n}\right).$$

Furthermore, we have that

$$\operatorname{rank}\left(\sum_{k=1}^{\ell} (c_k^{(\tau)} \mathbf{1}_n^\top) \circ S^{\circ k}\right) \le N_{\le \ell}(r) := \sum_{k=1}^{\ell} \binom{k+r-1}{r-1},$$

where $r = \operatorname{rank}(S)$. In particular, $F_1^{(L,\tau)}$ has at most $N_{\le \ell}(r)$ singular values that asymptotically separate from the bulk singular values of $F_0^{(L,\tau)}$.

2. No-spike regime. *If instead $\kappa > \frac{1}{2}$ and $|1 - \eta\lambda_{wd}| \le \rho_\star < 1$ for all sufficiently large n, then, with probability $1 - o(1)$,*

$$F_1^{(L,\tau)} = F_0^{(L,\tau)} + o\left(\sqrt{n}\right).$$

In particular, in this regime no finite-rank polynomial spike separates at the $\sqrt{n}$ scale.

Theorem 3.1 establishes $\kappa = \frac{1}{2} + \hat{\gamma} - \delta - 2\nu$ as the fundamental threshold parameter governing feature learning. We identify a subcritical regime ($0 < \kappa < \frac{1}{2}$) where feature learning occurs; this requires the network's effective "learning budget" to sit within the window $\hat{\gamma} < \delta + 2\nu < \frac{1}{2} + \hat{\gamma}$. The lower bound dictates that escaping the initial bulk spectrum requires a sufficiently large gradient step (δ), a sufficiently strong initial feature spike (ν), or a combination of both. If the budget is too small (i.e., $\delta + 2\nu < \hat{\gamma}$), then the network falls into the no-spike regime ($\kappa > \frac{1}{2}$). In this state, the network exhibits "lazy" training where the spectral update is overwhelmed by noise, and no new features are learned at the $\sqrt{n}$ scale.

We remark that Theorem 3.1 does not take into account the bulk covariance anistropy parameter α. We have observed experimentally that large values of α suppress most of the predicted spikes. We reason that the anisotropy and ill-conditioning of the bulk can lead to near-collinearity for some pairs of columns and thus the smaller singular values do not escape the bulk.

3.1 NTK vs. MF scaling

Under NTK-like scaling ($\hat{\gamma} = 0$), feature learning is theoretically permissible but fragile. The learning budget is given by $0 < \delta + 2\nu < \frac{1}{2}$; taking larger gradient steps or starting with a stronger spike causes the polynomial proxy to break down, leading to instability. Under MF-like scaling ($\hat{\gamma} = 1/2$), the window expands to $\frac{1}{2} < \delta + 2\nu < 1$. This higher theoretical ceiling allows the network to safely accommodate more aggressive gradient steps ($\delta \to 1$), which makes this setting more natural for feature learning to occur. This is in line with previously-established theory (Lee et al., 2019, Suzuki et al., 2023).

4 Gaussian equivalence

In this section, we derive a Gaussian equivalence (GE) property for the untrained features and analyze limiting test error. We do not develop this result for the trained feature matrix because GE is known to fail under large gradient steps (Ba et al., 2022); following the work of Moniri et al. (2024), we circumvent this difficulty by applying GE to the bulk term.

The GE property for kernel random matrices have been studied by Karoui (2010), Cheng and Singer (2013), Fan and Montanari (2019), etc. More recently, these equivalence results have been used in the study of random feature models (see, e.g., Goldt et al. (2022), Montanari and Saeed (2022), Mel and Pennington (2022), Hu and Lu (2023), etc.). Roughly speaking, this property asserts that the training and test errors of a random features model are asymptotically equivalent to that of a linear Gaussian model. GE is a particularly useful framework because it allows us to analyze highly nonlinear random matrices. In particular, the spectra of so-called information-plus-noise random matrices that are central to these results have been studied in Dozier and Silverstein (2007), Capitaine (2014), and Péché (2019).

4.1 Setup

For notational simplicity, let $F_0 = F_0^{(L,\tau)}$ and $F_1 = F_1^{(L,\tau)}$. Define

$$\hat{F}_1 \coloneqq \hat{F}_1^{(L,\tau)} = F_0 + \sum_{k=1}^{\ell} (c_k^{(\tau)} \mathbf{1}_n^\top) \circ S^{\circ k}.$$

This is the decomposition of the feature matrix that we will study from now on (which is asymptotically valid because F_1 and $\hat{F}_1$ are close in operator norm).

Since our network involves a scaling parameter in the form of γ_m, we will need to normalize objects such as the test error, ridge estimator, and the resolvent in order to avoid trivial scaling. We begin with the test error. For any $a, b \in \mathbb{R}^m$, define $b = \gamma_m a$ and the test error of b to be $\mathcal{L}_{\text{test}}(b) = \mathbb{E}_{f, y_{\text{test}}}[(y_{\text{test}} - f^\top b)^2 \mid \mathcal{F}_n]$, where $\mathcal{F}_n$ is the training sigma-field. Here, the expectation is taken over $(x_{\text{test}} y_{\text{test}})$, where $f = \sigma_L^\tau(W_1 x_{\text{test}})$ is the test feature vector with $x_{\text{test}} \sim \mathcal{N}(0, \Sigma)$ and $y_{\text{test}} = f_*(x) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. For fixed $\lambda_{\text{ridge}} > 0$, define the normalized ridge estimator

for the features to be

$$\begin{aligned}\hat{b}(F) = \hat{b}_{\gamma_m,\lambda_{\text{ridge}}}(F) &\coloneqq \arg\min_b \left\{ \frac{1}{n}\|y - Fb\|_2^2 + \lambda\|b\|_2^2 \right\} \\ &= \frac{1}{n}\left(\frac{1}{n}F^\top F + \lambda I\right)^{-1} F^\top y \\ &= \frac{1}{n}F^\top \left(\frac{1}{n}FF^\top + \lambda I\right)^{-1} y,\end{aligned}$$

where $\lambda = \lambda_{\text{ridge}}/\gamma_m^2 \in (0,\infty)$ for some $\lambda_{\text{ridge}} > 0$. Let $\hat{b}_0 = \hat{b}(F_0)$ and $\hat{b}_1 = \hat{b}(\hat{F}_1)$.

We also set up the following notation before we can state our result. For convenience, define the untrained feature matrix F_0 columnwise as $F_0 = [g_1, ..., g_m]$, where

$$F_{0,j} = g_j \coloneqq \sigma_L^\tau\Big((1-\eta\lambda_{\text{wd}})\tilde{X}_B w_j^{(0)}\Big) \in \mathbb{R}^n, \quad j \in [m].$$

We also define $\hat{F}_1$ columnwise. Define

$$\bar{\tau}_j^2 \coloneqq (1-\eta\lambda_{\text{wd}})^2 \big(w_j^{(0)}\big)^\top \hat{\Sigma} \big(w_j^{(0)}\big)$$

and

$$(c_{>1,L}^{\bar{\tau}_j})^2 \coloneqq \sum_{k=2}^{L} k! \bar{\tau}_j^{2k} (c_k^{\bar{\tau}_j})^2, \quad k \geq 2.$$

The scaled variance $\bar{\tau}_j^2$ are mainly defined to absorb $1-\eta\lambda_{\text{wd}}$ and simplify notation. We correspondingly define $c_{>1,L}^{\bar{\tau}_j}$; this sum of scaled Hermite coefficients represents the influence of the "nonlinear" terms in the finite Hermite expansion. This will be made more clear later on. Then, for some (normalized) ridge regularization $\lambda > 0$, define the normalized sample-covariance resolvent

$$R_0 \coloneqq \left(\frac{1}{n}F_0F_0^\top + \lambda I_n\right)^{-1}.$$

We use the Lindeberg exchange method (Lindeberg, 1922) to replace the columns of F_0 with a Gaussian surrogate $\tilde{F}_0$, i.e., the columns are defined as

$$\tilde{F}_{0,j} = \tilde{g}_j \coloneqq (1-\eta\lambda_{\text{wd}})c_1^{\bar{\tau}_j}\tilde{X}_B w_j^{(0)} + c_{>1,L}^{\bar{\tau}_j} z_j, \tag{2}$$

where Z is a random Gaussian matrix with columns $z_j \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_n)$, $j \in [m]$, and Z is independent of $(\tilde{X}_B, W_0)$. Roughly speaking, the surrogate linearizes the nonlinearity σ and essentially asserts that applying σ to each element of the features is equivalent to adding Gaussian noise. This is why GE theory has become widely used to study problems in high-dimensional statistics. We also define the corresponding normalized sample-covariance resolvent

$$\tilde{R}_0 \coloneqq \left(\frac{1}{n}\tilde{F}_0\tilde{F}_0^\top + \lambda I_n\right)^{-1}.$$

4.2 Main result

Theorem 4.1 (Y.–Sonthalia–Montúfar)**.** *Suppose Assumptions 1–2 of Sonthalia et al. (2025), Assumptions 1–2, and other technical conditions hold, with L and $\lambda > 0$ fixed. Then, the following hold jointly:*

1. *For each admissible matrix $A \in \mathbb{R}^{n\times n}$ with $\|A\|_2 = O(1)$,*

$$\frac{1}{n}\operatorname{tr}(AR_0) - \frac{1}{n}\operatorname{tr}(A\tilde{R}_0) \to_P 0.$$

2. *For each admissible vector pair $u, v \in \mathbb{R}^n$ satisfying $\|u\|_2, \|v\|_2 = O(\sqrt{n})$,*

$$\frac{1}{n}u^\top R_0 v - \frac{1}{n}u^\top \tilde{R}_0 v \to_P 0.$$

Proof sketch. We proceed as follows. Define $F^{(0)} \coloneqq F_0$ and $F^{(m)} \coloneqq \tilde{F}_0$. Construct matrices $F^{(j)} \in \mathbb{R}^{n\times m}$ by replacing the first j columns of $F^{(0)}$ with the first j columns of $\tilde{F}_0$, i.e.,

$$F^{(j)} = [\tilde{g}_1, ..., \tilde{g}_j, g_{j+1}, ..., g_m], \quad j \in [m].$$

Define the resolvent

$$R^{(j)} \coloneqq R(F^{(j)}) \coloneqq \left(\frac{1}{n}F^{(j)}F^{(j)\top} + \lambda I\right)^{-1}.$$

For a deterministic symmetric matrix $A \in \mathbb{R}^{n\times n}$ with $\|A\|_2 = O(1)$, define

$$\Phi_A(F) \coloneqq \frac{1}{n}\operatorname{tr}(AR(F)).$$

For deterministic vectors $u, v \in \mathbb{R}^n$ with $\|u\|_2, \|v\|_2 = O(\sqrt{n})$, define

$$\Psi_{u,v}(F) \coloneqq \frac{1}{n}u^\top R(F)v.$$

For each j, let

$$M_j \coloneqq [\tilde{g}_1, \ldots, \tilde{g}_{j-1}, g_{j+1}, \ldots, g_m] \in \mathbb{R}^{n\times(m-1)}.$$

Then, $F^{(j-1)}$ is obtained from M_j by inserting g_j as the j-th column, while $F^{(j)}$ is obtained from M_j by inserting $\tilde{g}_j$. Define the leave-one-out resolvent

$$S_j \coloneqq \left(\frac{1}{n}M_jM_j^\top + \lambda I_n\right)^{-1}.$$

For any functional H the deterministic telescoping identity gives

$$H(F^{(0)}) - H(F^{(m)}) = \sum_{j=1}^{m}\left[H(F^{(j-1)}) - H(F^{(j)})\right].$$

Taking $H = \Phi_A$ and $H = \Psi_{u,v}$, we obtain

$$\Phi_A(F_0) - \Phi_A(\tilde{F}_0) = \sum_{j=1}^{m}\left[\Phi_A(F^{(j-1)}) - \Phi_A(F^{(j)})\right]$$

and

$$\Psi_{u,v}(F_0) - \Psi_{u,v}(\tilde{F}_0) = \sum_{j=1}^{m}\left[\Psi_{u,v}(F^{(j-1)}) - \Psi_{u,v}(F^{(j)})\right],$$

respectively.

Fix j. Since

$$F^{(j-1)}F^{(j-1)\top} = M_jM_j^\top + g_jg_j^\top,$$

we have

$$R^{(j-1)} = \left(S_j^{-1} + \frac{1}{n}g_jg_j^\top\right)^{-1}.$$

Applying the Sherman–Morrison formula, we obtain

$$R^{(j-1)} = S_j - \frac{n^{-1}S_jg_jg_j^\top S_j}{1 + n^{-1}g_j^\top S_jg_j}, \quad R^{(j)} = S_j - \frac{n^{-1}S_j\tilde{g}_j\tilde{g}_j^\top S_j}{1 + n^{-1}\tilde{g}_j^\top S_j\tilde{g}_j}.$$

Subtracting $R^{(j)}$ from $R^{(j-1)}$ yields

$$R^{(j-1)} - R^{(j)} = \frac{1}{n}\left[\frac{S_j\tilde{g}_j\tilde{g}_j^\top S_j}{1 + n^{-1}\tilde{g}_j^\top S_j\tilde{g}_j} - \frac{S_jg_jg_j^\top S_j}{1 + n^{-1}g_j^\top S_jg_j}\right].$$

For the trace functional, define

$$\Theta_{j,A}(h) \coloneqq \frac{n^{-2}h^\top S_jAS_jh}{1 + n^{-1}h^\top S_jh}.$$

Then,

$$\Phi_A(F^{(j-1)}) - \Phi_A(F^{(j)}) = \Theta_{j,A}(\tilde{g}_j) - \Theta_{j,A}(g_j).$$

For the bilinear functional, define

$$\Theta_{j,u,v}(h) \coloneqq \frac{n^{-2}u^\top S_j h h^\top S_j v}{1 + n^{-1}h^\top S_j h}.$$

Then,

$$\Psi_{u,v}(F^{(j-1)}) - \Psi_{u,v}(F^{(j)}) = \Theta_{j,u,v}(\tilde{g}_j) - \Theta_{j,u,v}(g_j).$$

This reduces the problem to controlling these increments.

By Hölder's inequality and hypercontractive bounds on g_j and $\tilde{g}_j$, we get

$$\left|\mathbb{E}\left[\Theta_{j,A}(\tilde{g}_j) - \Theta_{j,A}(g_j)\right]\right| \le C\frac{o(1)}{n} + C\frac{(1-\eta\lambda_{\mathrm{wd}})^4}{n}.$$

and

$$\left|\mathbb{E}\left[\Theta_{j,u,v}(\tilde{g}_j) - \Theta_{j,u,v}(g_j)\right]\right| \le C\frac{o(1)}{n} + C\frac{(1-\eta\lambda_{\mathrm{wd}})^4}{n}.$$

Summing over $m \asymp n$ gives

$$\left|\mathbb{E}[\Phi_A(F_0) - \Phi_A(\tilde{F}_0)]\right| = o(1), \quad \left|\mathbb{E}[\Psi_{u,v}(F_0) - \Psi_{u,v}(\tilde{F}_0)]\right| = o(1).$$

Finally, we use the conditional Efron–Stein inequality to control the variance and prove concentration. □

Several remarks are in order. Part of the notion of *admissibility* involves working conditional on a certain high-probability event. This result also requires control of the (conditional) moments and covariance of the surrogates (i.e., matching first and second moments). Finally, fully understanding the test error actually requires a technical strengthening of Theorem 4.1 since the other objects that show up in the test error are more complicated than matrix products and bilinear forms.

Theorem 4.1 strengthens and generalizes similar results in several respects: 1) the bulk covariance is allowed to be anisotropic, 2) the matrix and vector probes (i.e., A, u, v in the theorem statement) can be dependent on the data bulk, and 3) the equivalence is at the level of the resolvent, which is more directly tied to ridge regression and thus generalization. We remark that this result also holds (and is easier to prove) for Gaussian weights. The uniform-spherical initialization in our setting is a (minor) complication that is treated with additional work. Finally, the ridge risk and resolvent limits of this surrogate can be analyzed using standard tools from random matrix theory (see, e.g., Bai and Silverstein (2010)).

4.3 Limiting test error

Our goal now is to quantify the difference between the test errors $\mathcal{L}_{\mathrm{test}} : \mathbb{R}^m \to \mathbb{R}$ using $\hat{b}_0$ and $\hat{b}_1$, i.e., ridge estimators using the untrained and trained features, respectively. Here, we suppose that Assumptions 3 and 4 to hold. It turns out that after a lot of algebra, this difference be expressed as the finite-dimensional identity

$$\mathcal{L}_{\mathrm{test}}(\hat{b}_0) - \mathcal{L}_{\mathrm{test}}(\hat{b}_1) = \mathcal{T}_n + 2\mathcal{S}_n^\top H^{-1}\mathcal{G}_n - \mathcal{S}_n^\top H^{-1}\mathcal{Q}_n H^{-1}\mathcal{S}_n - 2\left(\Xi_n + \mathcal{S}_n^\top H^{-1}\Omega_n\right), \quad (3)$$

where

$$\mathcal{S}_n \in \mathbb{R}^{2r_\ell}, \quad \mathcal{T}_n \in \mathbb{R}, \quad \mathcal{G}_n \in \mathbb{R}^{2r_\ell}, \quad \mathcal{Q}_n \in \mathbb{R}^{2r_\ell \times 2r_\ell},$$
$$\Xi_n \in \mathbb{R}, \quad \Omega_n \in \mathbb{R}^{2r_\ell}, \quad H \in \mathbb{R}^{2r_\ell \times 2r_\ell}$$

and we define $r_\ell \coloneqq N_{\le \ell}(r)$. Thus, the dimensions of these objects depend on the step size, which dictates the upper bound on the number of singular value spikes. Many of these objects are complicated and defining them explicitly is tedious. Instead, we opt to give a more high-level discussion of (3) and focus on how Theorem 4.1 fits into this machinery.

Each of these objects involve the resolvent R_0 in matrix products, matrix-vector products, or bilinear forms. The more complicated terms (e.g., $\mathcal{G}_n$ and $\mathcal{Q}_n$) involve sandwiched covariance terms. These

terms usually take the form $F_0\Sigma_0 F_0^\top$, where Σ_0 is the covariance of the untrained bulk test features. We use Theorem 4.1 (and its technical strengthening) to show that these objects jointly converge to deterministic limits, i.e.,

$$(\mathcal{S}_n, \mathcal{T}_n, \mathcal{G}_n, \mathcal{Q}_n, \Xi_n, \Omega_n) \to_P (\mathcal{S}, \mathcal{T}, \mathcal{G}, \mathcal{Q}, \Xi, \Omega).$$

Thus, the limiting test error becomes

$$\mathcal{L}_{\text{test}}(\hat{b}_0) - \mathcal{L}_{\text{test}}(\hat{b}_1) \to_P \Lambda_\ell \coloneqq \mathcal{T} + 2\mathcal{S}^\top H^{-1}\mathcal{G} - \mathcal{S}^\top H^{-1}\mathcal{Q}H^{-1}\mathcal{S} - 2\left(\Xi + \mathcal{S}^\top H^{-1}\Omega\right).$$

However, without stronger assumptions and structure, we cannot hope to reduce Λ_ℓ into nice constants (e.g., Hermite coefficients). What is known is that $\mathcal{G}_n, \mathcal{Q}_n, \Omega_n \to_P 0$, which greatly simplifies the computation for Λ_ℓ.

This result is analogous to Theorem 4.5 of Moniri et al. (2024). They consider isotropic Gaussian data and similarly derive a step size window corresponding to the number of terms ℓ in the Hermite expansion. However, due to this simpler setting, they were able to attain the following key insight: the difference in test errors Λ_ℓ is nonnegative. In particular, ℓ increasing as the step size increases implies a "staircase property" for the test error: for step sizes within a range that corresponds to a particular value of ℓ, the test error does not (meaningfully) decrease. The test error then drops sharply when the step size increases from the range defined by ℓ to the range defined by $\ell + 1$. This is in line with the theory established in prior work. For example, Abbe et al. (2021) prove that deep neural networks satisfying some conditions sequentially learn higher-order Fourier coefficients of structured Boolean functions. Moreover, Abbe et al. (2022) show how two-layer networks in the mean-field regime trained with stochastic gradient descent learn sparse functions on a hypercube. Berthier et al. (2025) study the problem using the language of dynamical systems to show that the empirical risk decreases non-monotonically (i.e., regions where the risk plateaus alternates with the risk sharply decreasing).

In our setting, we cannot guarantee that Λ_ℓ is strictly nonnegative. In fact, it may even be the case that some spikes cause the test error to increase. In order to make this tractable, we would most likely need to first consider step sizes in the window where $\ell = 1$. In this setting, we have closed-form expressions for each of the spike sources in the feature matrix, which then simplifies the terms in (3). We expect Λ_ℓ to then be much nicer analyze in this case. A negative result of this type is of theoretical interest and sheds light on how feature learning happens on a finer scale. Although it is known that neural networks can be both task-specific and data-adaptive, we do not yet have a comprehensive understanding of the mechanisms driving this phenomenon; this is a potential first step towards that goal.

5 Conclusions and discussion

5.1 Limitations

It can be argued that this one-gradient-step training procedure is just a form of weight initialization that promotes feature learning. In that respect, the network weights in this setting are not being trained. Moreover, the sample splitting procedure is only natural if one has pretrained models, but the analysis does not translate to the setting where feature learning and ridge regression use the same data. Finally, our model of bulk anisotropy is specific and may not extend well to more general settings; however, we expect that many of the broader implications of our work (e.g., with respect to the NTK and MF scalings) are still valid in practical settings.

5.2 Future work

While there has been extensive work studying the spectra of the network's weights and feature matrices at initialization (Damian et al., 2022, Adlam et al., 2022) as well as after one step of gradient descent (Cui et al., 2024), characterizing the spectral evolution across longer timescales remains challenging. However, progress has been made in the following respects. The recent work of Olsen et al. (2026), which established the first theoretical explanation for the heavy-tailed spectra of weight matrices observed in trained networks. Moreover, Bordelon and Pehlevan (2022) and Montanari and Urbani (2026) have characterized timescales where feature learning occurs via dynamical mean-field theory. Bringing these two lines of work together is of particular interest.

5.3 Contributions

C.N. worked on Sections 1, 2, and part of 3. E.Y. worked on Sections 3, 4, and 5. The slides were split equally.

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